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|---------------------------------|----------------------------------|---------------------|--------------------|
| <b>Course Name:</b>             | <b>Digital Design Laboratory</b> | <b>Semester:</b>    | <b>III</b>         |
| <b>Date of Performance:</b>     | <b>16 / 10/ 2023</b>             | <b>Batch No:</b>    | <b>B2</b>          |
| <b>Faculty Name:</b>            | <b>Bharath H. N.</b>             | <b>Roll No:</b>     | <b>16010122158</b> |
| <b>Faculty Sign &amp; Date:</b> |                                  | <b>Grade/Marks:</b> | ___/25             |

**Experiment No: 8**  
**Title: 1-bit adder on VHDL**

**Aim and Objective of the Experiment:**

To implement 1-bit adder on VHDL

**COs to be achieved:**

**CO4:** Implement digital networks using VHDL

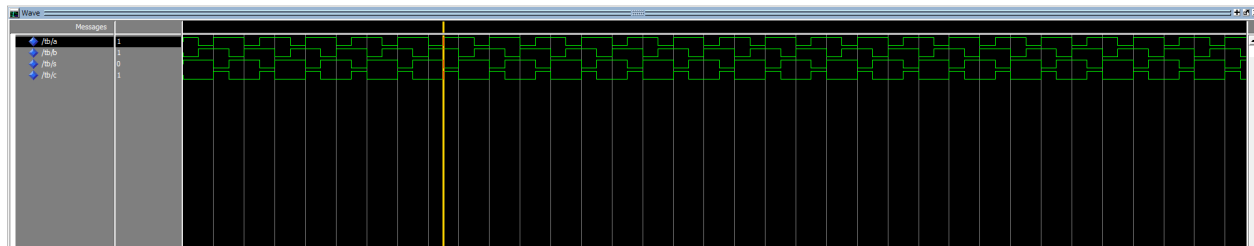
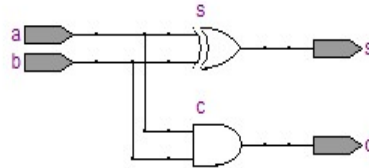
**Tools used:**

Quartus, ModelSim

**Theory:**

A 1-bit adder, a fundamental component of digital circuits, performs binary addition of two 1-bit numbers. It utilizes logic gates to generate the sum and carry-out outputs. A half-adder adds two bits without considering the carry from the previous stage, while a full-adder accounts for the carry input. Using VHDL, a hardware description language, the 1-bit adder can be designed as a combinational circuit. VHDL facilitates the creation of a structural and behavioral description of the adder. In practice, this simple unit serves as a building block for constructing larger multi-bit adders, enabling arithmetic operations in microprocessors and digital systems.

**Implementation Details**



Tips & Tricks

HA.vhd

```

1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity HA is
5    port(a,b:in std_logic;
6          s,c: out std_logic);
7  end HA;
8
9  architecture HA_arch of HA is
10 begin
11   s<= a xor b;
12   c<= a and b;
13 end HA_arch;

```

Tips & Tricks

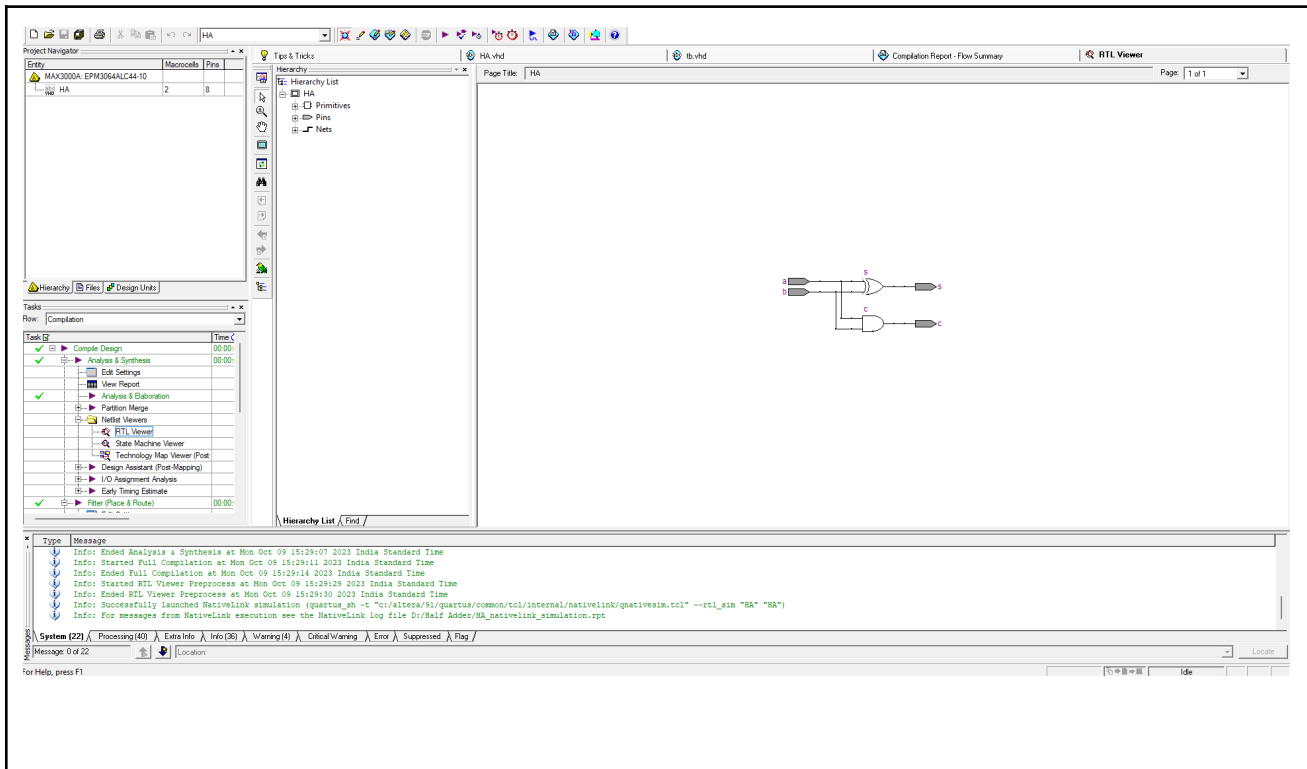
HA.vhd

tb.vhd

```

1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity tb is
5  end tb;
6
7  architecture tb_arch of tb is
8    component HA is
9      port(a,b:in std_logic;
10           s,c: out std_logic);
11    end component;
12    signal a,b,s,c:std_logic;
13    begin
14      tbm: HA port map(a,b,s,c);
15    process
16    begin
17      a<='1';
18      b<='0';
19      wait for 5ns;
20
21      a<='0';
22      b<='1';
23      wait for 5ns;
24
25      a<='1';
26      b<='1';
27      wait for 5 ns;
28    end process;
29  end tb_arch;
30
31

```



## Post Lab Subjective/Objective type Questions:

### 1. How can 1-bit adder be used to implement a 4-bit adder?

A 1-bit adder, also known as a half-adder, can be used as a building block to create a 4-bit adder. To build a 4-bit adder, you would cascade four 1-bit adders together, with each 1-bit adder handling one bit of the input and output.

**1. Define the Inputs and Outputs:** A 4-bit adder takes two 4-bit inputs, A and B, and produces a 4-bit sum (S) and a carry-out (Cout).

**2. Cascading Half-Adders:** To add two 4-bit numbers, you start by adding the least significant bits (LSBs) first. So, you connect the LSBs of A and B to the inputs of the first half-adder, resulting in a 1-bit sum (S0) and a carry-out (Cout0). The sum S0 is the LSB of the output, and Cout0 is then carried to the next stage.

**3. Repeat the Process:** Connect the next bit (bit 1) of A and B to the inputs of the second half-adder. Additionally, connect Cout0 from the first half-adder to one of the inputs of the second half-adder. This simulates adding bit 1 along with the carry-out from the previous stage. You get another 1-bit sum (S1) and carry-out (Cout1).

**4. Continue Cascading:** Continue this process for bits 2 and 3, connecting them to the inputs of the third and fourth half-adders and taking carry-outs from the previous stages. This will give you S2, S3, Cout2, and Cout3.

5. **Final Output:** The final 4-bit sum (S) is composed of S3, S2, S1, and S0 from the individual half-adders, while the carry-out Cout3 represents the carry-out of the entire 4-bit addition.

## 2. What is VHDL used for?

VHDL can be used for designing hardware and for creating test entities to verify the behavior of that hardware. VHDL is used as a design entry format by a variety of EDA tools, including synthesis tools such as Quartus Prime Integrated Synthesis, simulation tools, and formal verification tools.

## Conclusion:

Successfully implemented 1 bit adder on VHDL.

**Signature of faculty in-charge with Date:**